

The prescribed-parity induced-subgraph minimum on eleven vertices

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Abstract

For a graph with prescribed binary vertex labels, call a vertex set admissible if every vertex in its induced subgraph has degree of the prescribed parity. We determine the smallest possible maximum admissible order over all simple eleven-vertex graphs without isolated vertices and all labelings: it is four. The lower bound is a finite computational proof. Twelve Boolean formulas, one for each label weight, encode precisely the possible counterexamples; all twelve have retained, externally checked DRAT refutations. The upper bound is attained by the complement of a seven-cycle together with two disjoint edges, under an explicit labeling. We give the reduction, the construction, and instructions for replaying the certificates. Comparison is restricted to the supplied version of the source paper; priority is not asserted.

1 The parameter and its context

All graphs are finite, undirected, and simple. For a labeling $\ell : V(G) \rightarrow \mathbb{F}_2$, a set $S \subseteq V(G)$ is ℓ -admissible if

$$\deg_{G[S]}(v) \equiv \ell(v) \pmod{2} \quad (v \in S).$$

The empty set is admissible. Following Guo, Gutin, Hao, Shi, Zhang, and Zhou [1], define

$$h_\ell(G) = \max\{|S| : S \text{ is } \ell\text{-admissible}\}, \quad f_{\text{oe}}(G) = \min_{\ell: V(G) \rightarrow \mathbb{F}_2} h_\ell(G),$$

and, for $n \geq 2$, put

$$m_{\text{oe}}(n) = \min_{\substack{|V(G)|=n \\ \delta(G) \geq 1}} f_{\text{oe}}(G).$$

Disconnected graphs are included in this minimum.

Theorem 1. *One has $m_{\text{oe}}(11) = 4$. Thus every simple eleven-vertex graph without isolated vertices, with any binary labeling, has an admissible set of at least four vertices. For $G = \overline{C_7} \sqcup K_2 \sqcup K_2$, the labeling equal to one on $\overline{C_7}$ and zero on the other four vertices has $h_\ell(G) = 4$.*

Theorem 9 of the inspected arXiv version 1 of [1] gives $f_{\text{oe}}(G) \geq |V(G)|/6$, which yields only $m_{\text{oe}}(11) \geq 2$ after integer rounding. Its Proposition 31 establishes $f_{\text{oe}}(\overline{C_7}) = 2$, and the subsequent discussion states additivity over components. These facts give the upper construction used here. The discussion on page 14 reports $m_{\text{oe}}(n) = \lceil 2n/7 \rceil$ through $n = 10$, expressly describes those computer results as not independently verified apart from Proposition 31, and reports heuristic searches through order 18. The contribution of Theorem 1, relative to that version, is a complete checkable lower certificate at order eleven with the matching upper bound. We do not infer current open status or priority from this comparison. As adjacent computational methodology, Zeng [2] studies parity encodings and checked SAT certificates in the Erdős–Szekeres problem; that reference is not used to prove the present theorem.

2 The matching construction

Number the vertices of C_7 cyclically by $0, \dots, 6$, let $H = \overline{C_7}$, and label every vertex of H by one. If $T \subseteq V(H)$ is admissible, the handshake lemma implies that $|T|$ is even, so T is a proper subset of $V(C_7)$. For each $v \in T$,

$$\deg_{H[T]}(v) = |T| - 1 - \deg_{C_7[T]}(v). \quad (1)$$

Consequently every degree in $C_7[T]$ is even. A proper induced subgraph of a cycle is a disjoint union of paths. Any path with an edge has a degree-one endpoint; hence $C_7[T]$ has no edges. An independent set in C_7 has at most three vertices: mapping each selected vertex to its next cyclic neighbor gives an injection into the unselected vertices. It follows that the even integer $|T|$ is at most two. The pair $\{0, 2\}$ attains two in H .

Now adjoin edges $\{7, 8\}$ and $\{9, 10\}$ and label their four vertices zero. In each of these components an admissible set contains at most one vertex, and either singleton is admissible. For a disjoint union, a set is admissible exactly when its intersection with every component is admissible, since no degree receives a contribution from another component. Component maxima can therefore be attained simultaneously. The resulting graph-label pair has

$$h_\ell(G) = 2 + 1 + 1 = 4,$$

attained by $\{0, 2, 7, 9\}$. Its degree sequence is seven entries equal to four and four entries equal to one, so it has no isolated vertices. This proves $m_{\text{oe}}(11) \leq 4$ without a computational assumption.

3 The certified lower bound

Fix $V = \{0, \dots, 10\}$. We describe a formula F_w for each $w \in \{0, \dots, 11\}$, using the representative labeling

$$\ell_w(v) = \begin{cases} 1, & v < w, \\ 0, & v \geq w. \end{cases}$$

Every binary labeling of eleven vertices is taken to exactly one of these weight classes by a vertex permutation. The graph's edge assignment is permuted at the same time and remains unrestricted. Thus it suffices to rule out counterexamples for these twelve labelings; this is not a restriction to twelve labelings of one fixed graph.

Encoding graphs and inadmissible sets

Use the $\binom{11}{2} = 55$ Boolean variables $e_{uv} = e_{vu}$, $u < v$, for edges. The eleven clauses

$$\bigvee_{u \in V \setminus \{v\}} e_{uv} \quad (v \in V) \quad (2)$$

express exactly that no vertex is isolated. No connectivity constraint or further symmetry restriction is imposed. For each $S \subseteq V$ with $|S| \geq 4$ and $v \in S$, define the parity bit

$$p_{S,v} = \bigoplus_{u \in S \setminus \{v\}} e_{uv}.$$

It equals $\deg_{G[S]}(v) \bmod 2$. Let $L_{S,v}$ be the literal $p_{S,v}$ when $\ell_w(v) = 0$, and $\neg p_{S,v}$ when $\ell_w(v) = 1$. It is true exactly when the prescribed parity fails at v . The clause

$$\bigvee_{v \in S} L_{S,v} \quad (3)$$

therefore says precisely that S is inadmissible.

If $\sum_{v \in S} \ell_w(v)$ is odd, the handshake lemma already makes S inadmissible for every graph. Such sets need no clause. For every other set of size $4, 5, \dots, 11$, include (3). There are

$$\sum_{s=4}^{11} \binom{11}{s} = 1816$$

sets before these justified omissions. All sizes in this range are necessary for the reduction: admissibility need not be preserved by vertex deletion.

To obtain conjunctive normal form (CNF), chain two-input XOR gates to compute each $p_{S,v}$. A gate $c = a \oplus b$ is represented by

$$\begin{aligned} & (a \vee b \vee \neg c) \wedge (a \vee \neg b \vee c) \\ & \wedge (\neg a \vee b \vee c) \wedge (\neg a \vee \neg b \vee \neg c). \end{aligned} \tag{4}$$

Its eight-row truth table gives exactly $c = a \oplus b$. Each output uses a fresh auxiliary variable. Hence any assignment of the edge bits has a unique extension satisfying all gate clauses.

Lemma 2. *The formula F_w is satisfiable if and only if there is a simple graph G on V without isolated vertices with $h_{\ell_w}(G) \leq 3$.*

Proof. A satisfying assignment specifies a graph obeying (2). The forced parity bits and (3) make every retained set inadmissible; the handshake lemma excludes every omitted set. Thus no set of size at least four is admissible. Conversely, any such graph with $h_{\ell_w}(G) \leq 3$ assigns the edge bits and uniquely determines the gate outputs. The degree clauses hold, and every retained set has a parity mismatch, so all subset clauses hold as well. $\square$

Complete certificate coverage

CaDiCaL 3.0.1 produced binary DRAT refutations of all twelve formulas. The retained refutations were checked by the external program `drat-trim`; every branch returned exit code zero and the status `s VERIFIED`. A subsequent separate verification pass checked every CNF/proof pair again with the same checker. Table 1 gives the actual formula sizes and coverage.

The semantic verification directly parsed each actual DIMACS file. It checked all eleven degree clauses, reconstructed every required incident-edge parity chain, tested each XOR gate on all eight Boolean assignments, and checked the mismatch signs. It also required that every clause and auxiliary variable be consumed, excluding unchecked extra constraints. Across the twelve formulas it checked 285,864 gates on 2,286,912 truth-table rows and 11,804 subset clauses. In each branch the retained subsets plus the handshake-excluded subsets total exactly 1,816. This is complete symbolic verification of the encoding, not sampling of graph assignments.

The computational step uses the soundness of the executed DRAT checker: a verified refutation certifies unsatisfiability of its supplied CNF. Lemma 2 then excludes every counterexample of each weight. The permutation argument covers every graph-label pair, giving $m_{\text{oe}}(11) \geq 4$. Together with Section 2, this proves Theorem 1.

4 Reproducibility and limits

The accompanying computational dossier retains all twelve exact CNFs and binary DRAT proofs, the generators and verifiers, their logs, and the upper witness. Its manifest records the decisive paths and checksums; its reproduction instructions give exact replay commands. Python 3 and `drat-trim` suffice

Table 1: The twelve counterexample formulas. Variables include the 55 edge bits and all auxiliary bits. “Subset clauses” counts retained sets; each formula also has the eleven degree clauses and all XOR clauses.

Weight	Variables	Clauses	Subset clauses	DRAT check
0	44,627	180,115	1,816	Verified
1	17,625	71,139	848	Verified
2	22,825	91,975	884	Verified
3	22,351	90,099	904	Verified
4	22,359	90,139	912	Verified
5	22,349	90,099	912	Verified
6	22,333	90,031	908	Verified
7	22,323	89,987	904	Verified
8	22,331	90,019	904	Verified
9	22,369	90,179	912	Verified
10	22,449	90,519	932	Verified
11	22,583	91,091	968	Verified

for verification; CaDiCaL is needed only to regenerate proofs. Run in an extracted working copy, since the scripts write reports. All twelve checks must succeed; an interrupted check certifies no branch.

As a direct cross-check of the construction, a separate degree-counting program tested all $2^{11} = 2048$ vertex subsets of the upper witness. The counts of admissible sets, indexed by size $0, 1, \dots, 11$, were

$$(1, 4, 18, 56, 56, 0, 0, 0, 0, 0, 0, 0).$$

This is a finite computational proof with a conventional dependency on the proof checker and execution environment. The graph-theoretic reduction is explained above and was separately checked against the actual clauses; the whole statement has not been formalized in a proof assistant. There is no floating-point or probabilistic step. No conclusion for general n , no classification of extremizers, and no priority claim is made. The literature comparison uses the supplied arXiv version 1 of [1]; later versions and equivalent or subsequent results have not been verified.

Computational and AI assistance. OpenAI Codex assisted with the mathematical argument, generation and inspection of code and certificates, and preparation of the manuscript. A separate model context reconstructed the encoding, replayed all twelve proof checks, and checked every subset of the upper witness. This constitutes a separate verification pass, not external peer review or statistical independence. CaDiCaL generated the refutations and `drat-trim` checked them; model-generated assertions are not substitutes for those certificates or for the reduction in Section 3.

References

- [1] Q. Guo, G. Gutin, Y. Hao, Y. Shi, Y. Zhang, and Y. Zhou, *Bounds on Odd and Odd-Even Induced Subgraphs*, arXiv:2608.02339v1, 3 August 2026. <https://arxiv.org/abs/2608.02339v1>.
- [2] Zijian Zeng, *One-Sided Parity Compression for Certified SAT Attacks on the 33-Point Erdős–Szekeres Problem*, 2026. <https://doi.org/10.2139/ssrn.7334758>.